

Step 1: Group binary representation of the minterms according to the number of 1's contained, as shown in Table 3-5, column (a). This is done by grouping the minterms into five sections separated by horizontal lines. The first section contains the number with no 1's in it. The second section contains those numbers that have only one 1. The third, fourth and fifth sections contain those binary numbers with two, three and four 1's respectively. The decimal equivalents of the minterms are also carried along for identification.

Step 2: Any two minterms which differ from each other by only one variable, can be combined, and the unmatched variable removed. Two minterm numbers fit into this category if they both have the same bit value in all positions except one. The minterms in one section are compared with those of the next section down only, because two terms differing by more than one bit cannot match.

The resulting term, together with the decimal equivalents, is listed in column (b) of the table. The variable eliminated during the matching is denoted by a dash in its original position. In this case $m_0(0000)$ combines with $m_4(0001)$ to form $(000-)$. This combination is equivalent to the algebraic operation:

$$m_0 + m_4 = w'x'y'z' + w'x'y'z = w'x'y'$$

Minterm m_0 also combines with m_2 to form $(00-0)$ and with m_8 to form (-000) . The result of this comparison is entered into the first section of column (b). The minterms of sections two and three of column (a) are next compared to produce the terms listed in the second section of column (b). All other sections of (a) are similarly compared and subsequent sections formed in (b). This exhaustive comparing process results in the four sections of (b).

Step 3: The terms of column (b) have only three variables. A 1 under the variable means it is unprimed, a 0 means it is primed, and a dash means the variable is not included in the term. The searching and comparing process is repeated for the terms in column (b) to form the two-variable terms of column (c).